

Response to the R37 Referee Report

Manuscript: Neural Bellman Operators

Revision: R39, 7 October 2026

Report: review/econometrica-numerical-methods-r37-2026-10-07-9792d3d

Report commit: 71949aa40c62c960dab824137bed12bb3516ff85

Reviewed manuscript: 9792d3dee69351f672dcc09098422b35207060a7

Inherited new evidence: 2bb7a39a16c558da8269becc642b180847b2eb5c

We thank the referee for recognizing the earlier revision’s policy-comparison chain and for distinguishing mathematical correctness from a persuasive numerical contribution. We agree that an exact quadratic construction and a confounded timing contrast cannot alone substantiate the broader numerical claims. We respond by extending the original paper, not by replacing its topic with matrix factorization or deleting its economic applications.

The revised article adds a constructive nonlinear full-policy theorem, an all-state verifier for genuinely trained ReLU continuations, a residual-based native floating-point implementation theorem, complete-work accounting, identified factorial comparisons, and a nonlinear investment counterfactual with independently certified reoptimization gains. It retains the controlled-diffusion framework, recursive preferences, temporal selves, games, existing full-policy theory, historical article and supplement, and unfavorable conventional comparisons. Every active main-text and technical-supplement component is materialized into the current source tree. Historical sources remain unchanged.

There are three distinct types of evidence. The 315 economic services were already deposited in R38 and are re-audited here; they are not described as new R39 executions. The R39 breakpoint checks are new offline verification calculations, not new training runs or timing observations. The mathematical proofs state their assumptions separately from the finite regression checks. This response is to an owner-commissioned advisory report, not to an editorial decision by the Econometric Society.

B1 and M6: Nonquadratic continuation and a meaningful neural construction

The new section “Constructive Neural Bellman Operators beyond quadratic continuations” proves a residual-oscillation full-policy comparison for nonlinear finite-horizon recursive economies on compact state and action spaces. Its comparator is all feasible adapted policies, not the neural class or an action catalogue. The loss bound uses globally enclosing residuals, an actual selected-action allowance, and terminal error. The proof uses monotonicity and cash invariance and is independent of a Gaussian quadratic identity.

A constructive proposition verifies the assumptions for interval Bellman evaluation, continuous feasible actions, and off-grid states. Refining state and action meshes and arithmetic precision yields a finite procedure at any positive target for a fixed finite horizon. Its continuation has an explicit ReLU representation. We do not regard that representation alone as a neural computational advantage over the equivalent spline routine.

The deposited second implementation trains every affine hidden weight of a 31-unit ReLU network. Its own continuation selects actions; an independent interval bracket certifies those actual actions. Data generation, neural fitting, verification and policy evaluation all enter the cost. No convergence of unrestricted Adam or absence of representation bias is assumed. A certified constructed fallback

separates termination of the overall procedure from acceptance of every learned candidate.

The new breakpoint proposition further certifies the uniform difference between the trained continuation and its reference over every state, including nondyadic hidden-unit roots. All 60 stored date records map to 17 distinct networks. The new bounds are below 0.001, compared with roughly 0.027–0.031 under the prior mesh-Lipschitz check. Independent exact rational endpoint evaluation verifies all 17, with two additional edge-case fixtures. This sharpening is not relabeled as a new policy-loss or service-time result.

B2 and M1: Precision, caching and update-rule identification

The active evidence now uses fixed64/adaptive precision crossed with uncached/cached target inverses. All four cells share the same update rule, candidate-policy check boundary and economic stopping rule. An additional cached fixed method searches the ordered native set binary32, binary64; unsuccessful tuning trials are charged. The structural policy remains in the comparison.

The factorial isolates the precision and caching contrasts within that fixed update rule. It does not estimate an interaction with every historical update rule. Those older methods and their results are retained as separate historical comparisons, not pooled into this factorial. The introductory interpretation and historical-evidence preface explicitly withdraw a causal reading of the old joint adaptive-cached clock contrast.

The identified results are not uniformly favorable to adaptation. At target 1e-4, fixed64-cached and adaptive-cached have approximately 0.415 and 0.433 second median complete-service times. Each has eleven factor updates and three inverse builds. Their uncached counterparts build eleven inverses. The structural median is approximately 0.170 seconds. We preserve this result and locate the contribution in verified construction and policy accuracy rather than a timing victory unsupported by the design.

B3 and M2: Common economic targets and first-pass stopping

Five targets, from 1e-2 to 1e-10, are evaluated at the first successful economic certificate. Initial and intervening failed checks remain in the records and in total work. Tables report achieved policy error and time together; the supplement gives every cell and its counters.

First-pass stopping does not force a discrete quadratic update to meet the constraint at equality. At target 1e-6, the adaptive procedure still oversolves substantially. We report that slack instead of calling all five comparisons binding. The nonlinear construction supplies a separate nearly binding mesh frontier: first-pass bounds 0.9352, 0.4704, 0.2359 and 0.1181 for targets 1, 0.5, 0.25 and 0.125. Cumulative construction times include preceding coarser failures and are not confused with the full process clock or counted as independent observations within one repetition.

B4, M4 and M5: Realistic arithmetic and total work

The new residual-verification proposition treats arbitrary stored proposals Z and V. Enclosing MZ minus the original factor Gram and V minus the proposed matrix product bounds the complete deviation from the ideal factor step. Original stored inputs are used, so conversion errors, inverse approximation, caching, multiplication and final storage error are included. Proposals use binary32 or binary64 and verification uses outward binary64 balls. No exact rational matrix solve is used by these production services.

The proof covers every certified initial radius below one and explicitly distinguishes that range from the earlier half-radius budget. A direct terminal Gram certificate is distinguished from a squared-radius update. Failing all available native types is an explicit implementation failure, not evidence of impossibility at higher precision.

The complete arithmetic-operation statement includes target-inverse builds, every update attempt, residual verification, complete own-policy evaluations, full policy checks, actor solves and output work. It states storage, the dependence of contraction count on initial radius and tolerance, conditioning in the residual enclosure, and the entry of moment margins and actor/evaluation allowances into the economic allocation. Proposal multiply-add counters are not presented as complete certification-operation counters. Native factors remain stored in binary64; binary32 proposal work is not claimed to reduce persistent factor memory.

This is a native arithmetic bound, not an arbitrary-precision bit-complexity theorem. The policy certificate concerns the exact-real interpretation of stored feedback coefficients; floating-point actuator execution needs the separate execution allowance already in the manuscript. These distinctions are explicit in the theorem and implementation discussion.

B5, M3 and M7: Scale, repeated timing and transported failures

The inherited, source-frozen protocol has 315 isolated services, with three clocks per deterministic object and one separately recorded warm-up. The policy design includes dimensions 2, 4, 8, 16 and 32; horizons 4, 8 and 12; different condition parameters; an additional moment-margin case; the target sweep; and changes of scale 1.005, 1/4 and 4 under cold and transported starts. All 288 policy services certify their requested target.

The study has controlled thread counts and a fixed shuffled order, but no CPU affinity or frequency control. Medians and observed min-max ranges are reported, with startup-inclusive clocks retained separately. Three repetitions do not support population inference or a universal timing ranking. The finite grid does not establish asymptotic scaling or exhaustive near-singularity performance.

Transported starts are not restricted to successful gates. The small change takes three new adaptive factor updates rather than twelve from cold construction. Each large change fails all three transported factor gates and falls back. Warm setup is charged jointly, so the full warm service can be slower than a cold solve. Complete records retain these failures. Native precision fallbacks not observed in a particular economy are not invented to strengthen the narrative.

B6 and M8: Original scope, title and theorem boundaries

We retain “Neural Bellman Operators,” the original topic, and the complete economic framework. We respectfully do not follow the proposed replacement by a narrower linear-quadratic factorization paper. Instead, the new theorem and nonquadratic capital realization expand the established constructive chain while preserving the prior unbounded-state recursive-risk construction.

The abstract now states the actual nonlinear comparison, verified floating-point implementation and economic counterfactual. It does not suggest that the nonlinear compact-state theorem, or the earlier quadratic theorem, automatically establishes all assumptions in the diffusion, endogenous-preference, temporal-self and game applications. Those applications retain their own assumptions, statements and proofs. This is an extension of the existing paper, not deletion of its central program or an unsupported universal neural approximation theorem.

B7: Economic interpretation and a substantive counterfactual

The new realization includes irreversible bounded investment, nonlinear accumulation, clipping, quartic adjustment costs and a capital shortfall penalty. The cost of investment changes by a factor of four, rather than a small perturbation of a quadratic state matrix. Both risk-neutral and risk-sensitive specifications are evaluated.

A fixed-rule exposure proposition derives the marginal repricing burden as discounted investment exposure under risk-tilted transition weights. An exact accounting identity separates mechanical repricing from the gain due to reoptimization. Independent shock-tree intervals certify a positive gain from the new neural-selected rule over retaining the old conventional rule at each of four reported initial states, for both risk specifications. The new-price spline policy remains a stronger work comparator; the economic gain is not a neural-versus-spline speed or welfare superiority claim. The experiment is theoretical, not empirically calibrated, and the global policy bounds are displayed in actual cost units.

M9: Literature and novelty

The article retains its polar-iteration, matrix-function, risk-sensitive-control and approximate-dynamic-programming references. It adds a direct discussion of Carson and Higham's three-precision iterative-refinement analysis and Rump's verified-numerical-computation survey. The distinction between computed residuals and certified enclosures is central to the new implementation proposition. The novelty is located in the numerical-error-to-policy-loss connection and the implemented nonlinear construction, not in classical matrix or spline identities.

M10: Self-contained publication and preservation

The current main article and active technical supplement contain their active source text directly; neither requires following historical revision directories for its argument. A combined local bibliography and native Econometrica class are included. The complete applications companion and previous article and supplement remain available separately, with their exact source snapshots and identities retained. A preservation ledger records every inherited active component and its hash.

The release audit records the review, reviewed manuscript, inherited evidence and source freeze separately; validates all 315 result hashes and all frozen code hashes; includes the new endpoint verifier; and records native compilation. No old service is retimed or replaced. The current paper, supplement and this response are the objects submitted for the next advisory review. Editorial significance remains for the referee to assess; neither compilation nor a numerical certificate is represented as journal acceptance.